SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE NAME: Artificial Intelligence COURSE CODE: CSA1710

NAME: Preethi . R REG NO: 192511352

COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively.(BL3)

Assessment Tool Weightage: 50%

QUESTIONS: 2

Duration : 1 Hour
Total Marks:30

Annexure A

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Constraint-Based Problem Solving

Student 25: Preethi R | Register Number: 192511352

1. A hospital needs to assign 3 nurses (N1, N2, N3) to 3 wards (ICU, General, Emergency) such that:

   Constraints:

   i. N1 cannot be assigned to Emergency ward

   ii. N2 must be assigned before N3 (ICU < General < Emergency)

   iii. Only one nurse per ward

   iv. N3 cannot be assigned to ICU

   v. All nurses must be assigned exactly one ward

   Apply Backtracking Search to find a valid assignment, showing all intermediate steps and constraint checks. State the final valid schedule.

2. A cleaning robot operates in a 5x5 room-grid from Start (S) to a docking Goal (G), with a limited battery allowing at most 8 moves.

   Constraints:

   i. Movement allowed: Up, Down, Left, Right

   ii. Cost of each move = 1 (battery unit)

   iii. Cannot pass through furniture (obstacle) cells

   iv. The robot must reach G within the battery limit

Using the above constraints and Depth-Limited Search (with the battery limit as depth bound), show step-by-step node expansion and determine whether a valid path exists within the limit, along with its cost.

Code of Conduct:

I R. Preethi (192511352) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:
R. Preethi

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	25%	Clearly defines the problem and identifies all relevant constraints accurately	Identifies most constraints with minor gaps	Partial understanding; misses key constraints	Fails to identify constraints correctly
Constraint Analysis	25%	Analyzes constraints effectively and prioritizes them logically	Provides reasonable analysis with some prioritization	Limited analysis; weak prioritization	No meaningful analysis or prioritization
Solution Development	25%	Develops optimal and feasible solutions satisfying all constraints	Develops workable solutions with minor limitations	Solutions partially satisfy constraints	Solutions do not meet constraints
Application of Concepts	15%	Effectively applies appropriate concepts, methods, or tools	Applies concepts with minor errors	Limited application; requires guidance	Unable to apply relevant concepts
Justification	10%	Provides strong justification for decisions with logical reasoning	Provides justification with some clarity	Weak or unclear justification	No justification provided

CO 1 - ASSESSMENT - 2

Name: Preethi R
Regno: 192511352
Coursecode: CSA1720

CONSTRAINT - BASED PROBLEM SOLVING

1) Backtracking Search - Nurse Assignment

Given

→ Nurse : N_1, N_2, N_3

→ Wards : ICU < General < Emergency

Constraints

1. $N_1 \neq$ Emergency
2. N_2 must be assigned before N_3 (Ward order of N_2 < ward order of N_3)
3. Only one nurse per ward
4. $N_3 \neq$ ICU
5. Every nurse gets exactly one ward

State Space

possible assignment

Nurse	Possible wards
N_1	ICU, General
N_2	ICU, General, Emergency
N_3	General, Emergency

Backtracking procedure

→ Available wards : {ICU, General, Emergency}

Try :

N_1 → ICU

constraint check:

N₁ Cannot be emergency

only one nurse assigned

current assignment:

$N_1 \rightarrow ICU$

continue.

→ Remaining wards: {General, Emergency}

Try:

$N_2 \rightarrow$ General

Constraint check:

N_2 is before N_3

current Assignment:

$N_1 \rightarrow ICU$

$N_2 \rightarrow$ General

continue.

→ Remaining ward: {Emergency}

Try:

$N_3 \rightarrow$ emergency

Constraint check:

* N_3 is not ICU
* N_2 comes before N_3
* One nurse per ward

First valid Solution found

Nurse	ward
N_1	ICU
N_2	General
N_3	Emergency

Backtracking to Search for other Solution

1) Return to N_2

Try:

$N_2 \rightarrow$ Emergency

Remaining ward: General

Assign:

$N_3 \rightarrow$ General

Constraint Check:

* N_2 must be before N_3
* Emergency is after General
* Constraint Violated.
* Reject this branch.

2) Return to N_1

Try:

$N_1 \rightarrow$ General

Constraint check:

N_1 is not emergency

Current assignment:

$N_1 \rightarrow$ General

Assign N_2.

Remaining wards: {ICU, Emergency}

Try:

$N_2 \rightarrow$ ICU

Constraint Check:

possible because ICU comes before emergency

current Assignment:

$N_1 \rightarrow$ General

$N_2 \rightarrow$ ICU

Assign N_3.

Remaining ward: {Emergency}

Try:

$N_3 \rightarrow$ Emergency

Constraint check:

* N_3 is not ICU
* ICU comes before emergency
* All constraints satisfied.

Second valid Solution

Nurse	ward
N_1	General
N_2	ICU
N_3	Emergency

3) Return to N_1

Try:

$N_1 \rightarrow$ Emergency

Constraint Check:

* N_1 cannot be assigned to Emergency
* Constraint Violated.

Final valid Schedule

The first valid Schedule obtained using backtracking Search is:

Nurse	Assigned ward
N_1	ICU
N_2	General
N_3	Emergency

2. Problem Solving using depth-Limited Search (DLS)

Problem Statement

* A cleaning robot operates in a 5x5 room grid.
* The robot Starts from S and must reach the docking Station G.
* Battery Capacity allows only 8 moves.

Constraint:

1. Allowed movement:

→ up
→ down
→ Left
→ Right

2. Cost of every move = 1 battery unit

3. Robot cannot move through obstacle cells.

4. Goal must be reached within 8 moves.

Room Grid

S	.	x	.	.
.	.	x	.	.
.	.	.	x	.
x	x	.	.	.
.	.	.	.	G

where:

→ S = Start

→ G = Goal

→ x = obstacle

→ . = Free cell

depth limit : Maximum depth = 8

Depth 0

current node : (0, 0)

Robot is at Start, Expand.

possible move : Down → (1, 0)

Depth 1

current node : (1, 0)

Possible moves: up → Already Visited

Right → (1, 1)

Choose right.

Depth 2

current node: (1,1)

possible moves: Left → visited

down → (2,1)

Choose down.

Depth 3

Current node: (2,1)

Possible moves: Right → (2,2)

Chose right

Depth 4

Current node: (2,2)

Possible moves: Down → (3,2)

Chose down.

Depth 5

Current Node: (3,2)

Possible moves: down → (4,2)

Choose down.

Depth 6

Current node: (4,2)

Possible moves: Right → (4,3)

Depth 7

Current node: (4,3)

possible move: Right → (4,4)

Depth 8

Current node: (4,4)

Goal Reached. Search terminates Successfully

Cost calculation

Total cost = 8 Battery $\overset{6}{\text{units}}$

* Using Depth - limited Search, the robot successfully reaches the docking station within the battery limit.

* The algorithm expands node level by level up to specified depth and finds a valid path with a total cost of 8 battery unit.